# Heydar Soudani

Data Scientist · Uncertainty Estimation · LLM (Post-)Training · RAG & Agents

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Based in the Netherlands · Wrapping up PhD (expected 2027) · Open to full-time roles

## Summary

**Final-year PhD candidate in Computer Science** at Radboud University training, building and evaluating LLM systems for **retrieval-augmented generation (RAG)**, **deep research agents** and **uncertainty quantification**. At **Thomson Reuters Labs**, built agentic search for legal research and a controller for deep research agents based on trajectory-level signals and analysis (EMNLP 2026); at **KPN**, brought uncertainty estimation to a customer-facing RAG chatbot. Hands-on with LLM fine-tuning (full, LoRA / QLoRA), RL post-training, synthetic data generation and benchmark construction. Published at ACL, EMNLP, COLING, SIGIR, SIGIR-AP, ICTIR and ACM Computing Surveys.

## Experience

**Thomson Reuters Labs** *Nov 2025 – May 2026*

*Data Scientist Intern (Applied AI Research)* Zug, Switzerland (Remote)

- **Agentic search for legal research**: replaced a single-step retrieval pipeline with a modular **agentic search system** (planner, query decomposer, retriever, reranker, summarizer) to raise recall. Worked with the **product team** on **prompt engineering** and component wiring, evaluating each change in the product.
- **Stagnant reasoning** (*EMNLP 2026*): showed across **7 deep research agents** and 3 datasets that recall plateaus after a few search calls while agents keep searching. Built **RAAC**, a controller that uses 4 unsupervised trajectory signals to continue, intervene or stop: 14 fewer search calls on average and up to 10% higher accuracy on BrowseComp-Plus, and 58% fewer iterations with higher recall on LegalSearch.
- **Query decomposition & reformulation study**: benchmarked decomposition and reformulation methods (HyDE, Query2doc, Rewrite-Retrieve-Read, ThinkQE) with **8 rerankers** (MonoT5, RankLLaMA, Rank1, Rank-R1, Qwen3, RankGPT, RankZephyr) and **rank fusion** (RRF, interleaving, bandits). Found that one-shot methods are capped by the LLM's own knowledge, which motivated agentic search.
- **Infrastructure**: **AWS** (SageMaker multi-GPU jobs, Bedrock), **Langfuse** trajectory tracing, Claude and GPT APIs, and open-weight Qwen models served with **vLLM**.

**KPN** *May 2025 – Oct 2025*

*Data Scientist Intern* Amsterdam, the Netherlands

- **Uncertainty for a production chatbot**: applied the axiomatic uncertainty method (*ACL 2025*) to KPN's customer-facing **RAG chatbot**, which answers from a **knowledge graph** and text, to score its confidence in each answer and flag unreliable responses.

**Radboud University** *Feb 2023 – Present*

*PhD Candidate, Trustworthy Generative Information Access* Nijmegen, the Netherlands

- **Uncertainty for search agents** (*SIGIR 2026*): built **R2C**, a **multi-sampling** uncertainty method that perturbs an agent's reasoning-and-search trajectory and scores how consistent its answers stay. Across 5 agents: +5% AUROC. Tested the uncertainty scores on downstream tasks: **abstention** (about 5% better F1) and **model selection** (about 7% higher exact match).
- **Uncertainty in RAG** (*ACL 2025*): for the **retrieve-then-generate** RAG setup, defined **5 axioms** for how uncertainty should change under retrieval, showed existing methods meet only 2, and derived a **calibration function** that brought RAG AUROC to or above the no-retrieval level.
- **Multimodal uncertainty** (*ICTIR 2026*): co-developed **LeMUQ**, a **fine-tuned RoBERTa** scorer based on **input perturbation**: it reads a VLM's token probabilities with the image or retrieved context removed, improving AUROC by 0.038 on average.
- **Fine-tuning vs. RAG** (*SIGIR-AP 2024*): to find whether RAG or fine-tuning better injects **long-tail, less popular knowledge** into LLMs, compared **full fine-tuning**, **QLoRA**, synthetic QA generation and RAG on **12 LMs** (80M–11B). RAG won on long-tail entities, and the proposed **Stimulus RAG** beat fine-tuning with no training cost.
- **Total Recall QA** (*SIGIR 2026*): co-built a **verifiable benchmark** for deep research agents where answering a query

requires finding **all** relevant documents (total recall), with about 2.3K queries over a 57.7M-document corpus. It remains challenging for state-of-the-art agents.

- **Synthetic conversational data** (*ACM Computing Surveys 2026*): led a survey of **synthetic multi-turn dialogue generation** for open-domain, task-oriented and information-seeking systems. Proposed a general framework covering seed data creation, utterance generation and **quality filtering**.
- **Tutorials**: Data Augmentation for Conversational AI, 2 editions (*CIKM 2023*, *WebConf 2024*).

**Atrovan** *Dec 2017 – Jun 2022*
*Web Developer (part-time)* Tehran, Iran

- **Products**: built and maintained **4 production web products** for industrial and individual users: smart metering, fleet management, home automation and building management.
- **Engineering**: **React.js**, JavaScript, **REST APIs**, **PostgreSQL**, **Docker**, **Kubernetes** and Git, in a 10+ person **Scrum** team, contributing to **system design**.

## Selected Publications

[1] **EMNLP 2026**. When Deep Research Agents Stagnate: Enhancing Reasoning with Retrieval-Aware Agent Control
[2] **SIGIR 2026**. Uncertainty Quantification for Retrieval-Augmented Reasoning
[3] **SIGIR 2026**. Total Recall QA: A Verifiable Evaluation Suite for Deep Research Agents
[4] **ICTIR 2026**. Uncertainty Quantification for Multimodal Retrieval Augmented Generation
[5] **ACM Computing Surveys 2026**. A Survey on Recent Advances in Conversational Data Generation
[6] **ACL 2025**. Why Uncertainty Estimation Methods Fall Short in RAG: An Axiomatic Analysis
[7] **SIGIR-AP 2024**. Fine Tuning vs. Retrieval Augmented Generation for Less Popular Knowledge
[8] **COLING 2022**. Persian Natural Language Inference: A Meta-learning Approach

## Technical Skills

**Languages:** Python, SQL, Bash, JavaScript, C/C++
**ML / DL:** PyTorch, Hugging Face (Transformers, PEFT, Datasets), scikit-learn, TensorFlow, NumPy, pandas
**Uncertainty & evaluation:** uncertainty quantification (multi-sampling, input perturbation), calibration (AUROC), abstention & model selection, LLM-as-judge, benchmark construction, retrieval & trajectory analysis
**LLM training & post-training:** SFT, full fine-tuning, LoRA / QLoRA, RL post-training (GRPO, PPO, DPO, RLVR), reward design, synthetic data generation
**Agents & RAG:** deep research agents (ReAct, Search-R1, Tongyi-DR), agentic search, tool use, RAG over text and knowledge graphs, query decomposition & reformulation, dense & sparse retrieval (BM25, SPLADE, E5, Qwen3-Embedding), LLM rerankers, rank fusion, FAISS, LangChain / LangGraph, prompt engineering
**LLM inference & models:** vLLM, LiteLLM, OpenRouter; Claude, GPT, Qwen, GLM, GPT-oss, vision-language models
**MLOps & cloud:** AWS (SageMaker, Bedrock), multi-GPU training, HPC / Slurm, Langfuse, W&B, Docker, Kubernetes
**Software engineering:** Git, Linux, REST APIs, PostgreSQL, system design, Claude Code, Cursor

## Education

**PhD Computer Science** — Radboud University, 2023–2027 (expected)
**MSc Computer Engineering** — Sharif University of Technology, 2022
**BSc Electrical Engineering** — Polytechnic of Tehran, 2019
**BSc Computer Engineering** (minor) — Polytechnic of Tehran, 2018

## References

**Dr. Faegheh Hasibi** — Assistant Professor, Radboud University, Nijmegen · f.hasibi@cs.ru.nl
**Dr. Hamed Zamani** — Associate Professor, University of Massachusetts Amherst · zamani@cs.umass.edu
**Dr. Navid Rekabsaz** — Research Scientist, Grubel (formerly Thomson Reuters Labs) · navid.rekabsaz@gmail.com